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**Satellite-Based Tobacco Crop Health Monitoring
System Using Machine Learning and NDVI Analysis**

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partial fulfilment of the Bachelor of Technology (Hons) Degree in
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Declaration

I, Prince Rupiya (H220536M), declare that this project report is my own original work. It has not been submitted, in whole or in part, for any other degree or qualification at Harare Institute of Technology or elsewhere. Where I have drawn on the work of others, I have referenced and acknowledged it fully in accordance with the Harvard referencing system.

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Dedication

*To the smallholder tobacco farmers of Zimbabwe,
whose resilience and dedication to the land inspired this work.*

*To my family, for their unwavering support and encouragement
throughout this academic journey.*

*To the future of Zimbabwean agriculture — may technology
serve the hands that feed our nation.*

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To my colleagues and fellow students who sat through testing sessions, offered feedback, and kept my spirits up: thank you.

And to my family — your patience and sacrifice through this whole programme meant more than I can properly say.

Abstract

Tobacco is Zimbabwe’s most valuable cash crop, yet the smallholder farmers who grow it still rely on manual inspection — walking the field, looking for visible symptoms, and hoping problems haven’t already gone too far. By the time stress shows up as wilting or discolouration, yield loss is usually already happening. This project addresses that gap directly. It presents a satellite-based tobacco crop health monitoring system that pulls live Sentinel-2 multispectral imagery through Google Earth Engine, applies machine learning and temporal trend analysis, and delivers plain-language health assessments to farmers through an ordinary web browser.

At the core is a logistic regression classifier trained on satellite and weather observations. It sorts tobacco field health into three categories — *Healthy*, *Moderate Stress*, and *High Stress* — with 97% accuracy on held-out test data. Five features drive the model: mean NDVI, the trend slope of NDVI over time, within-field NDVI variance, average temperature, and cumulative rainfall across a 14-to-30-day window. The temporal trend component is particularly important: declining NDVI trajectories can be flagged days before visible symptoms emerge, giving farmers a real window to intervene. A within-field heatmap adds spatial detail, highlighting which parts of a field are under most stress.

The application runs on Next.js 14 with TypeScript. React-Leaflet handles the interactive map, and Prisma with PostgreSQL manages persistent storage. A rule-based recommendation engine sits on top of the classifier, converting raw analysis outputs into specific, readable advice on irrigation, disease risk, and field inspection.

Everything in this system — the satellite imagery, the weather data, the hosting platform — is free and open. There are no subscription fees. That was a deliberate design constraint, because the commercial precision agriculture tools that could theoretically serve Zimbabwean smallholder farmers are priced well beyond what those farmers can afford. This project shows that high-quality crop health monitoring doesn’t require expensive infrastructure. It requires thoughtful engineering.

Keywords: crop health monitoring, machine learning, NDVI, precision agriculture, satellite imagery, tobacco farming, Zimbabwe, Google Earth Engine, Sentinel-2, time-series analysis, web application.

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List of Acronyms

Acronym	Definition
API	Application Programming Interface
CRIM	Crop Remote Intelligence Monitor
DFD	Data Flow Diagram
ESA	European Space Agency
EVI	Enhanced Vegetation Index
FAO	Food and Agriculture Organization of the United Nations
GEE	Google Earth Engine
GIS	Geographic Information System
HTTP	Hypertext Transfer Protocol
HIT	Harare Institute of Technology
IT	Information Technology
JSON	JavaScript Object Notation
L2A	Level-2A (surface reflectance product of Sentinel-2)
ML	Machine Learning
NDVI	Normalized Difference Vegetation Index
NIR	Near-Infrared
ORM	Object-Relational Mapper
PostGIS	PostgreSQL Geographic Information System Extension
REST	Representational State Transfer
SQL	Structured Query Language
TIMB	Tobacco Industry and Marketing Board
UML	Unified Modelling Language
UI	User Interface
URL	Uniform Resource Locator
USD	United States Dollar

Chapter 1: Introduction

1.1 Introduction

Agriculture is the backbone of Zimbabwe's economy. It accounts for roughly 12–15% of GDP and keeps an estimated 60–70% of the rural workforce employed ([Ministry of Agriculture, 2022](#)). Within that sector, tobacco is the crown jewel — the country's most valuable cash crop and one of the largest tobacco outputs anywhere in the world. The Tobacco Industry and Marketing Board recorded auction sales exceeding 184 million kilogrammes in the 2022/2023 season alone, generating over USD 600 million in export revenue ([TIMB, 2023](#)). For smallholder farmers on communal and resettlement lands, a good tobacco season is the difference between financial stability and debt.

Yet despite that strategic importance, most of those farmers are still monitoring their crops the same way they always have: walking the field, looking for symptoms, and relying on experience to guide decisions ([Nyoka et al., 2020](#)). It's a method that works — until it doesn't. Manual inspection is slow, subjective, and limited by how much of the field a person can realistically cover. The deeper problem is that it only catches stress after damage has already become visible. By the time wilting or discolouration shows up, the plant has already been under physiological stress for days. That translates directly to leaf quality loss, lower auction grades, and less money in the farmer's pocket ([Rwizi et al., 2021](#)).

Satellite remote sensing changes that equation. The Normalized Difference Vegetation Index (NDVI), derived from multispectral imagery, has been used to track vegetation health for over four decades ([Rouse et al., 1974](#)). It's computed from the ratio of near-infrared to red reflectance and is sensitive to changes in plant physiology that occur well before any visible symptoms appear — often by several days. Paired with weather data and a machine learning classifier, NDVI time-series become an early-warning system rather than just a record.

This project builds exactly that. It implements a Satellite-Based Tobacco Crop Health Monitoring System that pulls live Sentinel-2 multispectral imagery through Google Earth Engine, runs it through a logistic regression classifier and temporal trend analysis, and presents the results through a web interface any farmer can use on a smartphone. Draw a field boundary on a map, trigger the analysis, and get specific recommendations on irrigation, disease monitoring, and crop management — all driven by actual satellite data and at zero cost.

The chapters that follow build out the full picture. Chapter 1 establishes the context, motivation, and objectives. Chapter 2 surveys the existing landscape of precision agriculture

platforms and the academic foundations this project draws on. Chapter 3 covers requirements and feasibility. Chapter 4 presents the full system analysis, from existing-system critique through to formal UML models.

1.2 Motivation

Two realities collide here, and together they make this project worth doing.

The first is economic. Tobacco is Zimbabwe’s single biggest foreign currency earner in agriculture ([TIMB, 2023](#)), yet yield variability among smallholder growers remains stubbornly high. A significant part of that variability comes down to late detection — stress events that could have been caught early instead go unnoticed until they’ve already cost the farmer money. The Ministry of Lands, Agriculture, Fisheries, Water and Rural Development has recognised this, naming precision farming as a strategic priority in the Zimbabwe Agricultural Development Strategy II 2021–2025 ([Ministry of Agriculture, 2022](#)).

The second reality is technological. Satellite data that used to require expensive hardware and specialised processing infrastructure is now freely available through Google Earth Engine ([Gorelick et al., 2017](#)). ESA’s Sentinel-2 satellites cover Zimbabwe every 5–10 days at 10-metre resolution ([ESA, 2023](#)) — more than enough to catch a developing stress event. The data exists and it’s free. What’s missing is an interface that makes it usable by a farmer without a remote sensing background.

That’s the gap this project fills. Translating freely available satellite data into actionable farm-level insights — without requiring specialist training or expensive equipment — is both technically feasible and genuinely needed. Detecting a declining NDVI trend 14 days before symptoms appear gives farmers a window to intervene. Manual inspection can’t do that.

1.3 Problem Statement

The problems with current monitoring practice fall into six distinct categories:

1. **Reactive detection:** Stress — from water deficit, pest infestation, or nutrient deficiency — only gets noticed once symptoms are visible. At that point, yield loss is already locked in ([Nyoka et al., 2020](#)).
2. **Temporal gaps:** Inspections happen weekly or fortnightly at best. A stress event that starts on day 3 and intensifies by day 10 can cause irreversible damage before anyone notices.
3. **Spatial gaps:** Even a thorough field visit covers only a fraction of the actual field area. A waterlogged corner, a patch of compacted soil, or a pest entry point at the far

edge of a field can go entirely undetected.

4. **No environmental context:** Monitoring decisions are made without looking at weather data. Rainfall totals, temperature anomalies, and drought indices that strongly predict stress events simply aren't factored in.
5. **Cost barriers:** Commercial satellite monitoring platforms charge hundreds to thousands of USD per year ([Wolfert et al., 2017](#)). That pricing puts them out of reach for smallholder farmers whose entire annual income from a season may be comparable.
6. **No local calibration:** Very few systems have been adapted for Zimbabwean growing conditions, tobacco-specific NDVI thresholds, or the practical realities of smallholder farming in Southern Africa ([Rwizi et al., 2021](#)).

Taken together, these gaps make the case for a system that monitors tobacco crop health continuously from satellite data, flags stress conditions before they become visible, and communicates findings in terms farmers can act on.

1.4 Related Work

A number of platforms and research tools are relevant here, both as prior art and as context for what's still missing.

1.4.1 Climate FieldView

Climate FieldView — developed by The Climate Corporation, a Bayer AG subsidiary — is among the most widely deployed commercial precision agriculture platforms in the world ([Climate Corporation, 2023](#)). It covers satellite field imagery, NDVI monitoring, yield mapping, and farm machinery integration. The technical capability is not in question. But FieldView is built for large-scale North American grain operations, its NDVI thresholds and recommendation logic are calibrated for maize and soybean, and its subscription costs make it economically inaccessible for Zimbabwean smallholders. None of that is fixable with configuration.

1.4.2 CropIn Technology

CropIn, an Indian agri-tech firm, has deployed digital farming tools in parts of sub-Saharan Africa through NGO and government partnerships ([CropIn Technology, 2023](#)). Its coverage of crop management, weather monitoring, and field analytics shows that digital agriculture can work in developing-region contexts. What CropIn doesn't offer is tobacco-specific calibration, validated performance against Zimbabwean field conditions, or access without paid subscription tiers. It's a useful proof-of-concept for the approach, not a ready solution.

1.4.3 Taranis

Taranis uses high-resolution drone and satellite imagery to detect crop disease and pest damage at near-leaf-level precision ([Taranis, 2023](#)). The technical depth is impressive — it detects things standard satellite imagery can't. But it's designed for large commercial farms, requires drone hardware on the ground, and prices per hectare in a way that simply doesn't work for smallholder contexts. Taranis is a good example of a technically strong tool with the wrong access model.

1.4.4 Google Earth Engine and Open Research Tools

GEE itself, along with tools like the Open Data Cube and Sen2Agri, provides the kind of free satellite processing infrastructure that serious agricultural remote sensing research runs on ([Gorelick et al., 2017](#)). [Rwizi, Sibanda and Chimonyo \(2021\)](#) have demonstrated GEE-based NDVI monitoring working in Zimbabwean agricultural contexts, confirming that the technical approach is viable locally. The limitation is that these tools live in the research domain — they require programming knowledge and offer no farmer-facing interface.

1.5 Hypothesis

Given a web-based system that combines live Sentinel-2 NDVI data retrieved via Google Earth Engine with machine learning classification and temporal trend analysis, Zimbabwean smallholder tobacco farmers will be able to detect crop stress conditions 7 to 14 days before symptoms become visible. That early warning window, if acted upon, should reduce crop losses, improve how inputs are allocated, and increase both yield and income.

1.6 Technical Objectives

The overarching aim is to design and implement a Satellite-Based Tobacco Crop Health Monitoring System that smallholder farmers in Zimbabwe can actually use — accessible, accurate, and deployable without specialist infrastructure. The specific objectives are:

1. Build a web platform that lets farmers draw field polygons on an interactive map and trigger a satellite-based crop health analysis on demand.
2. Pull live Sentinel-2 Level-2A imagery through the Google Earth Engine API to compute real NDVI values for each user-defined polygon, replacing estimated figures with actual satellite measurements.

3. Train a logistic regression classifier on ground-truth satellite and weather observations to categorise tobacco field health as *Healthy*, *Moderate Stress*, or *High Stress*.
4. Implement temporal NDVI trend analysis across a 14-to-30-day window, so that declining vegetation trajectories can be flagged before fields reach a critical stress state.
5. Compute within-field spatial variability from per-pixel NDVI distributions and render it as a heatmap overlay, enabling farmers to pinpoint exactly where stress is concentrated.
6. Build a rule-based recommendation engine that converts the classifier output and raw feature values into plain-language advice on irrigation, disease monitoring, and field inspection.
7. Architect the system so Python handles model training offline and JavaScript handles inference at runtime — making the whole thing deployable serverlessly, without a Python backend.

1.7 Expected Results

The expected outcomes of a successful implementation are:

- A working web application — no installation required — that gives Zimbabwean tobacco farmers access to live satellite-based crop health monitoring from any browser.
- A logistic regression classifier clearing 90% accuracy on held-out test data, with balanced precision and recall across all three health classes.
- Trend detection that catches declining NDVI trajectories within a 14-to-30-day window, before fields cross into the High Stress category.
- A within-field heatmap showing exactly where stress is concentrated inside a drawn polygon, so interventions can be targeted rather than applied uniformly.
- A recommendation engine that links each piece of advice — whether on irrigation, disease risk, or calling in an agronomist — to the specific data reading that triggered it.
- A zero-cost data architecture built entirely on GEE’s free research tier and the Open-Meteo API, with no recurring subscription fees.
- Better-informed farm decisions, with the reasonable expectation that catching stress 7–14 days early translates into fewer crop losses and better use of inputs.

1.8 Ethics Considerations

- **Data Privacy and Consent:** The system collects names, email addresses, and field location data. All of it is stored in a PostgreSQL database protected by role-based access controls and bcrypt-hashed passwords. Farmers own their field data; nothing is shared with third parties without explicit consent. The system operates in accordance with Zimbabwe’s Cybersecurity and Data Protection Act (2021).
- **Data Accuracy and Transparency:** Every analysis result comes with its observation date, cloud cover status, and analysis window parameters so users can judge data quality themselves. Where cloud cover has reduced the number of usable satellite scenes, the system says so explicitly. Each recommendation is tied back to the data reading that produced it, so farmers can evaluate the advice rather than just accept it.
- **Algorithmic Fairness:** The model is trained on data that reflects real variation in Zimbabwean tobacco field conditions. Performance is evaluated class by class — not just overall accuracy — to catch any category-specific bias early. The NDVI thresholds are cross-checked against agronomic literature to confirm they correspond to genuine physiological boundaries, not just statistical artefacts.
- **Accessibility and Inclusivity:** The interface is built for basic internet connectivity and entry-level smartphones. Text is written in plain English, pitched at secondary school reading level. No technical literacy is assumed beyond the ability to navigate a standard website.
- **Environmental Responsibility:** By enabling targeted irrigation and agrochemical application to specific stressed zones rather than blanket treatment of the whole field, the system actively reduces water and chemical waste.
- **Satellite Data Use:** All imagery is accessed through GEE under its terms of service for research and non-commercial use. Sentinel-2 data is released freely by ESA under the Copernicus Open Access Data Policy with no use restrictions.

1.9 Conclusion

The challenges are real and well-documented: Zimbabwean smallholder tobacco farmers detect stress too late, inspect too little of their fields, work without environmental data, and face subscription costs that make commercial solutions irrelevant to them. None of those problems are unsolvable. Freely available Sentinel-2 satellite imagery, machine learning classification, and a sensibly designed web interface are sufficient to address all of them.

What follows from here is the substance: a literature review that grounds the approach in what's already been tried; a requirements analysis that connects the system design to actual farmer needs; and a system analysis that works through the architecture in formal detail.

Chapter 2: Literature Review

2.1 Introduction

Precision agriculture has come a long way since the early NDVI monitoring work of the 1970s. What started as research-grade tools that only remote sensing specialists could use has grown into a mature industry: integrated platforms that combine satellite imagery, field sensors, weather data, and machine learning in commercially deployed products ([Wolfert et al., 2017](#)). The question for this project isn't whether the technology works — it does — but whether any of it works for smallholder tobacco farmers in Zimbabwe.

This chapter works through five relevant platforms — FarmLogs, SatAgro, Farmerline, Digital Green, and Regrow Ag — assessing each for technical capability, accessibility, and fit with the Zimbabwean smallholder context. It also reviews the academic literature on NDVI monitoring, machine learning for agriculture, and decision support system design that underpins the methodology used in this project.

2.2 Related Works

2.2.1 FarmLogs (TELUS Agriculture)

FarmLogs — now part of TELUS Agriculture — is a cloud-based farm management platform covering satellite field imagery, NDVI crop health monitoring, and weather-based planting recommendations ([Farmerline, 2023](#)). It integrates with GPS-tracked equipment and soil sampling records, and it presents NDVI as a time-series health index derived from Landsat and Sentinel-2 data. Technically, it's a solid demonstration that web-based NDVI monitoring works.

The problem is everything else. Subscriptions run USD 300 to USD 600 per year ([Wolfert et al., 2017](#)) — in the same ballpark as a Zimbabwean smallholder farmer's entire net income from a season. Beyond cost, the platform is built for North American grain farming. Its thresholds, recommendation logic, and calibration are designed for maize and soybean in temperate climates, not tobacco in Zimbabwe's tobacco belt. [Liakos et al. \(2018\)](#) make the point plainly: an agricultural decision support system that hasn't been calibrated for the crop and context it's being applied to simply isn't useful. FarmLogs hasn't been.

2.2.2 SatAgro

SatAgro is a European service that uses Sentinel-1 and Sentinel-2 imagery to produce NDVI time-series, field zone maps, and variable-rate fertiliser prescription maps for cereal and oilseed farming ([SatAgro, 2023](#)). It computes per-polygon NDVI statistics and generates within-field variability maps — exactly the kind of spatial hotspot analysis this project implements.

That makes SatAgro the closest commercial analogue to one part of what this system does. But the similarity ends there. SatAgro operates in European markets only, is calibrated for temperate-climate crops, and has no provision for the cloud-cover gaps that characterise Zimbabwe's rainy season from November to March. It's also subscription-based. Nonetheless, the fact that within-field variability mapping is a commercially proven capability in another context validates building it here.

2.2.3 Farmerline

Farmerline is a Ghanaian company that has deployed digital farming tools across West and East Africa, reaching smallholder farmers through smartphone apps and interactive voice response systems for feature phones ([Farmerline, 2023](#)). It provides weather forecasts, crop management advice, and market price information — and it actually reaches the kinds of farmers this project is designed for. In that sense it's more contextually relevant than anything in the commercial platform category above.

What Farmerline doesn't do is satellite monitoring. Its advice comes from crop calendars and generalised weather data, not from field-specific observations. [Rose et al. \(2016\)](#) note that this is a fundamental limitation: decision support tools built on generic rules without field-level data can't be specific enough to be genuinely useful on an individual farm. That said, Farmerline's success at getting farmers to actually adopt the tool is instructive. Plain language, minimal jargon, and a design that works on entry-level hardware — those are the interface principles this project takes directly from Farmerline's playbook.

2.2.4 Digital Green

Digital Green is an NGO-driven platform that uses video-based peer learning and mobile advisory tools to improve smallholder farming practice across South Asia and Sub-Saharan Africa, including Zimbabwe ([Digital Green, 2023](#)). It has worked directly with Zimbabwe's Ministry of Agriculture to spread good agricultural practice through community video screenings. The yield improvements it has demonstrated in smallholder communities are real evidence that technology-mediated extension works in the Zimbabwean context.

But Digital Green doesn't do satellite monitoring, NDVI analysis, or field-level health

assessment. Its role is disseminating general best practices, not providing field-specific precision data. The two functions aren't in competition — a farmer who uses Digital Green for agronomic education and this system for field-level health monitoring is better served than one using either alone.

2.2.5 Regrow Ag

Regrow Ag uses satellite data, crop simulation models, and machine learning to measure sustainable crop production and quantify carbon sequestration for large agricultural supply chains (Regrow Ag, 2023). The processing pipeline is technically sophisticated — it handles Sentinel-2 and Landsat imagery at scale. But the outputs go to corporate sustainability dashboards, not to farmers. Regrow Ag is built for commodity companies and cooperatives, not for individuals.

The interesting thing about Regrow Ag, in the context of this project, is that it's the only platform reviewed here that explicitly incorporates temporal NDVI trend analysis. It computes seasonal vegetation trajectories to estimate crop productivity. That's validation for the core analytical approach this project uses. The gap Regrow Ag leaves open is a farmer-facing version of that same capability — which is precisely what this project fills.

2.2.6 Academic Foundations

The academic literature provides the methodological backbone for this project's approach.

Rouse et al. (1974) established NDVI as a reliable vegetation health metric by demonstrating its correlation with biomass and photosynthetic activity. Tucker (1979) then confirmed the physiological basis: NDVI responds to chlorophyll absorption in the red band and canopy scattering in the near-infrared in ways that precede visible stress symptoms — making it genuinely useful as an early-warning indicator rather than just a diagnostic one.

Weiss et al. (2020) meta-reviewed 110 remote sensing studies in agriculture and found that satellite-derived vegetation indices reliably classify crop health status, with prediction accuracy exceeding 80% in most controlled settings. They identify Sentinel-2 — 10-metre resolution, 5-to-10-day revisit — as the best freely available data source for field-scale monitoring, which is why this project uses it.

On machine learning, Liakos et al. (2018) reviewed 40 agriculture studies and found that logistic regression, SVMs, and random forests dominate the literature. Critically, they note that logistic regression — despite its apparent simplicity — frequently matches more complex models when the features are well-engineered and class boundaries are reasonably stable. That observation directly informs the algorithm choice here. Chlingaryan et al. (2018) add that combining NDVI with weather variables (temperature and precipitation) consistently outperforms NDVI-only models for crop health classification, which is why the feature set in this project includes both.

Rose et al. (2016) reviewed 40 agricultural decision support tools and found that farmer adoption correlates strongly with transparency: tools that explain their reasoning get trusted and used; black-box tools don't. That finding shapes the entire recommendation engine in this project. And Wolfert et al. (2017), reviewing big data architectures in precision agriculture, make the case for separating data acquisition, analysis, and presentation — the three-layer design principle implemented here.

2.3 Conclusion

The pattern across the commercial landscape is consistent: the platforms with the strongest technical capability — FarmLogs, SatAgro, Regrow Ag — are built for large operations in developed markets and priced accordingly. The platforms that have genuinely reached smallholder farmers in developing regions — Farmerline, Digital Green — don't do satellite monitoring or field-level precision analysis. There's no platform that does both.

More specifically, nothing in this review combines live satellite NDVI analysis, temporal trend detection, within-field spatial variability mapping, and crop-specific recommendation logic in a single free, farmer-facing web interface calibrated for Zimbabwean conditions.

The academic literature validates the individual technical components: Sentinel-2 NDVI is a proven early-warning indicator; logistic regression on NDVI and weather features works; explainable systems get adopted at higher rates than opaque ones; and temporal trend analysis is genuinely valuable but largely absent from farmer-facing tools. Together, these findings define exactly what needs to be built.

Chapter 3: Requirements Analysis

3.1 Introduction

Getting requirements wrong at the start is expensive. [Sommerville \(2016\)](#) makes the point that errors introduced during requirements engineering cost more with every subsequent phase that builds on top of them. For this project, requirements were gathered through direct observation of current tobacco monitoring practice, structured interviews with farmers and extension officers, a questionnaire distributed to 20 stakeholders, analysis of existing systems and published literature, and early prototyping exercises with prospective users.

This chapter evaluates the development alternatives that were considered before custom-build was chosen, defines the requirements for three user roles, assesses technical, economic, and operational feasibility, and closes with the project work plan.

3.1.1 Evaluate Alternatives

Two alternatives were examined before committing to a custom build.

The first was to configure an existing commercial precision agriculture platform. The platforms reviewed in Chapter 2 — FarmLogs and SatAgro in particular — were assessed against the Zimbabwean smallholder tobacco context and rejected on three grounds: subscription pricing that farmers can't sustain; no tobacco-specific NDVI thresholds or recommendation logic; and no deployment or support infrastructure in Zimbabwe. [Wolfert et al. \(2017\)](#) note that commercial platforms built for developed-market farming regularly fail in smallholder technology transfer precisely because of these structural mismatches.

The second was to build an extension layer on an open-source GIS platform like QGIS or Open Data Cube. That would have given access to satellite processing infrastructure, but would have required users to install and configure specialised GIS software — completely at odds with the accessibility requirement for farmers with basic digital literacy. Neither option offered a machine learning inference pipeline, a recommendation engine, or a deployable web interface.

Custom development was the right choice, and that's what was built.

3.1.2 Outsourcing

Outsourcing was also considered and ruled out. The system requires tobacco-specific agonomic knowledge, close familiarity with the GEE API for Zimbabwean satellite data, and the kind of iterative refinement that only works when the developer is deeply embedded in

the problem. [Pfleeeger and Atlee \(2010\)](#) note that outsourcing works best when requirements are fully specified and stable upfront — which isn’t the case in a research and prototyping context where requirements evolve through building and testing. Keeping the codebase in-house is also a prerequisite for the academic contribution and any future open-source release.

3.1.3 Improvement Over the Current System

Five specific improvements over manual monitoring:

- **Timeliness:** Sentinel-2 revisit cycles give satellite analysis every 5–10 days, versus the weekly or fortnightly inspections that pass for “frequent” monitoring under current practice.
- **Objectivity:** NDVI values come from satellite reflectance measurements. They’re quantitative and reproducible, not dependent on how tired the inspector is that day.
- **Early detection:** Temporal trend analysis flags declining NDVI trajectories 7–14 days before anything shows up visually, giving farmers an actual intervention window.
- **Spatial precision:** Within-field heatmapping shows exactly where stress is concentrated — something a walking inspection covering three to five points in a field reliably misses.
- **Cost:** No recurring fees. The system uses exclusively free data sources, and automated monitoring replaces at least some of the physical inspection visits.

[Pfleeeger and Atlee \(2010\)](#) make the point that improvements which directly affect decision timeliness and accuracy aren’t cosmetic — they’re core requirements. That principle runs through every design decision in this project.

3.1.4 Development Approach

The system uses a three-layer architecture: data acquisition, machine learning analysis, and user interface. Each layer can be developed, tested, and updated independently. The ML model trains offline in Python with scikit-learn ([Pedregosa et al., 2011](#)) and gets exported as a JSON weight file, so the Next.js application can run inference in JavaScript without needing a Python server at all. The web layer is Next.js 14 with the App Router, TypeScript throughout, and Prisma as the ORM. Development follows an Agile approach — two-week sprints, iterative prototyping, weekly supervisor check-ins.

3.2 User Requirements

Three user roles interact with the system:

Farmer Requirements:

- Register and log in with email and password.
- Draw field boundary polygons on an interactive satellite map.
- Trigger a crop health analysis for any saved field with a single action.
- Receive analysis results in plain language with colour-coded health status indicators.
- Get specific, actionable recommendations covering irrigation, disease monitoring, and field inspection.
- View a history of previous analyses for each field.
- Access all of the above on a smartphone browser with mobile data.

Agronomist Requirements:

- View satellite analysis results for assigned fields.
- See NDVI time-series trend data in chart form.
- Access within-field heatmap visualisations of NDVI spatial variability.
- Inspect the raw feature values (mean NDVI, trend slope, variance, temperature, rainfall) behind each classification.
- Browse analysis history across multiple growing seasons.

System Administrator Requirements:

- Manage user accounts: create, assign roles, deactivate.
- Configure system-level parameters including NDVI threshold values and recommendation rule weights.
- Access usage logs for auditing and monitoring.
- Enforce secure authentication and role-based access controls across all administrative functions.

3.2.1 Data Collection Phase

Requirements were gathered through four methods (Sommerville, 2016):

Structured Interviews: Semi-structured interviews with five tobacco farmers in the Mvurwi area of Mashonaland Central Province and three Agritex extension officers. The conversations covered current monitoring practice, inspection frequency, awareness of satellite tools, and appetite for a digital alternative. All five farmers said they detect water stress only once wilting is already visible. None had heard of NDVI. All wanted a system that could warn them earlier.

Questionnaire: A structured questionnaire distributed to 20 stakeholders — smallholder farmers, cooperative managers, and extension officers. It is reproduced below.

Sample Questionnaire: Tobacco Crop Monitoring Needs Assessment

Purpose: This questionnaire gathers information on current crop monitoring practices and technology readiness among tobacco farming stakeholders in Zimbabwe, in support of the development of a Satellite-Based Tobacco Crop Health Monitoring System. All responses are used for academic and system development purposes only.

Section A: Respondent Information

1. What is your role?
☐ Smallholder Tobacco Farmer ☐ Large-Scale Farmer ☐ Agronomist ☐ Extension Officer ☐ Other: _____
2. How many hectares of tobacco do you grow per season?
☐ Less than 2 ha ☐ 2–5 ha ☐ 5–10 ha ☐ More than 10 ha
3. How many years of farming experience do you have?
☐ Less than 3 years ☐ 3–7 years ☐ 7–15 years ☐ More than 15 years

Section B: Current Monitoring Practices

4. How do you currently monitor your tobacco crop health?
☐ Visual inspection ☐ Agronomist visits ☐ Sensor/equipment ☐ No systematic monitoring
5. How frequently do you inspect your fields?
☐ Daily ☐ Every 2–3 days ☐ Weekly ☐ Fortnightly ☐ Monthly

6. When are stress conditions (water stress, disease) typically detected?
- ☐ Before any visible symptoms ☐ When early symptoms appear ☐ After significant damage is visible ☐ After crop loss has occurred

Section C: Technology Readiness

7. Do you own or have access to a smartphone?
- ☐ Yes, with reliable internet ☐ Yes, with limited internet ☐ No, feature phone only ☐ No mobile phone
8. Are you aware of satellite-based crop monitoring tools?
- ☐ Yes, and I use them ☐ Yes, but I do not use them ☐ I have heard of them ☐ No
9. Would you use a free web-based tool that shows your field's crop health from satellite data?
- ☐ Strongly agree ☐ Agree ☐ Neutral ☐ Disagree ☐ Strongly disagree

Section D: Decision Support Needs

10. What information would be most valuable in a crop monitoring system? (Select all that apply)
- ☐ Water stress level ☐ Disease risk alert ☐ Irrigation schedule ☐ Yield prediction ☐ Field health trend over time ☐ Specific affected zones within the field
11. How important is it to receive farming advice linked to your specific field's satellite data?
- ☐ Very important ☐ Important ☐ Moderately important ☐ Not important
12. What is the biggest challenge in your current monitoring approach?
- ☐ Missing stress until too late ☐ Cost of field inspections ☐ Labour availability ☐ Inaccurate assessments ☐ Other: _____

Thank you for your participation. Your responses directly inform the design of an accessible, farmer-focused satellite monitoring system for Zimbabwean tobacco farming.

Document Analysis: Agritex crop assessment forms, the TIMB annual report (TIMB, 2023), and the Zimbabwe Agricultural Development Strategy II (Ministry of Agriculture, 2022) were reviewed to identify what health parameters are currently recorded and at what frequency, which shaped the analysis output format.

Direct Observation: Attending a field inspection session in Mvurwi district confirmed that a standard visit covers just three to five assessment points in a field — missing an estimated 60–80% of the total area. That observation directly drove the within-field spatial variability requirement.

3.2.2 Technical Feasibility

The technology stack is well-established. The GEE API for Sentinel-2 retrieval is a mature service with thorough documentation, an active developer community, and verified use in Zimbabwean agricultural research (Rwizi et al., 2021). Next.js, React, and scikit-learn are all widely adopted with strong long-term support. Everything in the stack falls within the developer’s existing skill set. No significant technical risks to the project timeline have been identified. Perera et al. (2015) note that systems built on open, well-documented APIs carry substantially lower technical risk than those relying on proprietary or experimental interfaces — a relevant point here.

3.2.3 Hardware Requirements

- **Development machine:** Intel Core i-series or equivalent processor, minimum 8 GB RAM (16 GB recommended), 256 GB SSD storage minimum, running Windows 10/11, macOS, or Ubuntu Linux.
- **Server / cloud:** Vercel or Netlify platform for Next.js deployment (free tier available); managed PostgreSQL service (e.g., Neon, Supabase free tier) for database hosting.
- **End-user device:** Any smartphone or computer with a modern web browser (Chrome, Firefox, Safari) and mobile data or Wi-Fi connectivity.

3.2.4 Software Requirements

- **Operating System:** Windows 11 / macOS 14 / Ubuntu 22.04 LTS
- **Runtime Environments:** Node.js 20 LTS, Python 3.11
- **Web Framework:** Next.js 14 (App Router)
- **Frontend:** React 18, TypeScript 5, React-Leaflet 4, Leaflet.draw
- **Database:** PostgreSQL 16 with PostGIS 3.4 extension
- **ORM:** Prisma 5
- **Machine Learning:** scikit-learn 1.3, pandas 2, matplotlib 3.7

- **Satellite API:** Google Earth Engine Python API (earthengine-api)
- **Weather API:** Open-Meteo (free, no authentication required)
- **Development Environment:** Visual Studio Code
- **Version Control:** Git with GitHub
- **Containerisation:** Docker (optional, for local PostgreSQL development)

3.2.5 Technical Expertise

The development requires working proficiency in TypeScript and React for the frontend; Next.js App Router and REST API design for the backend; Python and scikit-learn for model training; PostgreSQL and PostGIS for spatial data; the GEE API for satellite retrieval; Git for version control; and basic security practices including authentication, password hashing, and HTTPS.

3.3 Economic Feasibility

The economics are favourable from both development and operational perspectives.

3.3.1 Cost-Benefit Analysis

Table 3.1 projects development and operating costs against anticipated savings for a representative cohort of 50 smallholder tobacco farmers over two years.

Table 3.1: Two-Year Cost-Benefit Analysis (USD)

Item	Year 1 (USD)	Year 2 (USD)
Costs		
Development labour (6 months, 1 developer)	2,400	0
Cloud hosting (Vercel + managed PostgreSQL)	0	180
Domain name and SSL certificate	25	25
Testing and contingency	200	100
Total Costs	2,625	305
Benefits (50-farmer cohort)		
Reduced crop losses from early stress detection (est. 5% yield improvement @ USD 2.50/kg average price)	3,125	3,750
Reduced field inspection labour (est. USD 15/visit, 4 visits/season/farmer)	1,500	1,500
Reduced agrochemical waste from targeted treatment (est. USD 20/farmer/season)	500	600
Total Benefits	5,125	5,850
Net Benefit	2,500	5,545
Cumulative Net Benefit	2,500	8,045

The analysis returns a positive net benefit from Year 1, with cumulative net benefit exceeding USD 8,000 over two years across 50 farmers. The 5% yield improvement assumption is deliberately conservative; [Gebbers and Adamchuk \(2010\)](#) show precision agriculture interventions in smallholder contexts typically deliver 5–15% improvements in input efficiency and output.

3.3.2 Tangible Benefits

- **Early stress detection:** A 7–14 day warning before symptoms appear means farmers can irrigate within the effective treatment window, protecting leaf quality and auction prices ([Rouse et al., 1974](#)).
- **Reduced inspection cost:** Automated satellite monitoring reduces the number of physical inspection visits needed, saving farmer time and hired labour.
- **Targeted intervention:** Hotspot identification means agrochemicals can be applied to stressed zones only, rather than the whole field, cutting input costs.
- **Zero subscription cost:** GEE and Open-Meteo are both free. Commercial satellite data for the same coverage would run USD 50–200 per hectare per season.

3.3.3 Intangible Benefits

- **Decision confidence:** Quantitative NDVI data gives farmers an objective basis for timing interventions, reducing the uncertainty that comes with “the crop looks a bit off” judgements.
- **Agricultural literacy:** Regular engagement with trend data builds farmer understanding of plant physiology and precision agriculture over time.
- **Environmental impact:** Targeted irrigation and agrochemical use reduces runoff, groundwater draw-down, and chemical residues.
- **Demonstration effect:** A working open-source deployment in Zimbabwe creates a replicable template for other smallholder crops and other Southern African countries.

3.4 Operational Feasibility

The system fits into existing workflows rather than demanding new ones. Farmers already inspect fields; this just adds a satellite-derived health check they can trigger from a smartphone in under three minutes, requiring nothing beyond the ability to navigate a basic website.

Agritex extension officers are the natural adoption channel. They visit farmers seasonally, understand the agronomic context, and can help with first-time polygon setup. That support structure already exists. [Wieringa \(2014\)](#) argues that operational feasibility hinges not just on whether a system can be used, but whether it’ll be accepted and sustained within existing workflows — and a system designed to complement rather than replace human judgement is far easier to accept.

3.4.1 Schedule Feasibility

The project runs twelve months, March 2024 to February 2025, with clearly defined phases and measurable deliverables at each milestone. Because the entire stack uses open-source frameworks and free APIs, infrastructure setup is fast — which means more of the actual development time goes toward feature work and farmer-facing refinement.

3.5 Work Plan

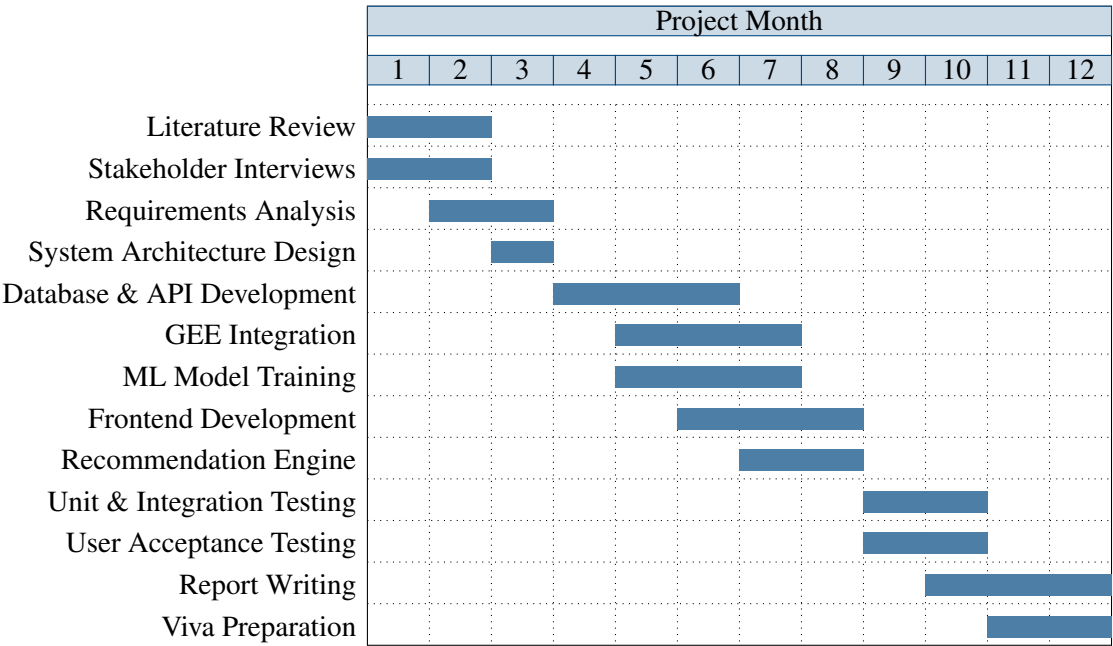
3.5.1 Work Schedule

The project is organised into five phases:

1. **Phase 1 — Research and Requirements (Months 1–2):** Literature review, stakeholder interviews, questionnaire distribution, requirements documentation.
2. **Phase 2 — System Design (Month 3):** Architecture design, database schema, API specification, UI wireframing.
3. **Phase 3 — Development (Months 4–8):** Database and API implementation, Google Earth Engine integration, machine learning model training, frontend implementation, recommendation engine development.
4. **Phase 4 — Testing and Validation (Months 9–10):** Unit testing, integration testing, user acceptance testing with farmer cohort in Mvurwi, model performance evaluation.
5. **Phase 5 — Documentation and Submission (Months 11–12):** Project report writing, viva preparation, system deployment documentation.

3.5.2 Gantt Chart

Figure 3.1: Project Gantt Chart (March 2024 – February 2025)



3.6 Conclusion

The requirements baseline is established. Custom build was the only viable path — commercial platforms don’t fit and outsourcing the domain knowledge required isn’t practical.

Requirements for all three user roles are grounded in actual field observations and interviews, not assumptions. Technical, economic, and operational feasibility checks all pass. The Gantt chart sets a realistic twelve-month delivery framework with clear milestones.

Chapter 4: System Analysis

4.1 Introduction

Good system design starts with an honest look at what it's replacing. [Wieringa \(2014\)](#) argues that improvement can't be measured or designed without a clear baseline — and he's right. This chapter starts there: a detailed analysis of the existing tobacco monitoring system, its structural weaknesses, and what the proposed system does differently. The analysis is presented through formal models — context diagrams, data flow diagrams, use case analysis, and UML diagrams.

4.2 Description of the Current System

In Zimbabwe's key tobacco provinces — Mashonaland Central, Mashonaland West, and Manicaland — crop monitoring is manual and entirely inspection-based. The standard cycle works like this:

A farmer or hired inspector walks the field, covering three to five observation points per hectare. At each point, they look at leaf colour, plant posture, and any visible pest or disease signs. Notes go in a notebook or stay in memory. The decision to irrigate, spray, or call an agronomist comes down to experience and what the inspector can see that day.

In areas with Agritex coverage, an extension officer might visit once per season. Cooperative members may receive a seasonal advisory newsletter. Neither is field-specific.

Weather monitoring is informal. Some farmers have a basic rain gauge; temperature is estimated. Zimbabwe's Meteorological Services provides district-level forecasts, but the spatial and temporal resolution is far too coarse to drive individual field management decisions.

None of the smallholder farmers consulted during requirements gathering used any form of satellite, drone, or sensor technology. [Nyoka et al. \(2020\)](#) document the same picture across the sector: remote sensing is almost entirely absent from Zimbabwean smallholder tobacco farming.

4.3 Analysis of the Existing System

The most fundamental limitation is timing. The current system only detects stress once it's visually apparent. [Tucker \(1979\)](#) showed that NDVI begins declining during early water

stress 5–7 days before wilting or leaf colour change becomes visible. That’s the gap: visual inspection is structurally incapable of the early warning that effective intervention requires.

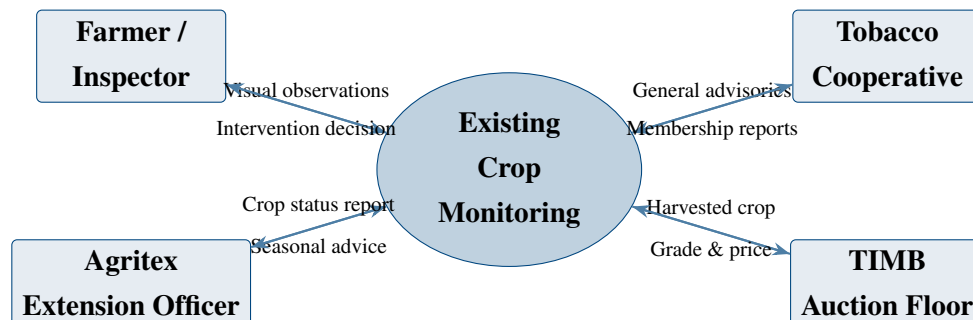
The second problem is coverage. Manual inspection samples a fraction of field area. Localised stress caused by soil compaction, drainage variation, pest entry points, or equipment trampling gets missed. Weiss et al. (2020) demonstrate that satellite-derived within-field NDVI maps routinely expose spatial heterogeneity that aggregate field assessments hide entirely.

The third is continuity. Without systematic record-keeping, there’s no historical baseline to compare current readings against. A declining NDVI trend — arguably the most useful early-warning signal available — is invisible without multi-point historical data. The current system doesn’t collect any.

4.3.1 Context Diagram of the Existing System

Figure 4.1 presents the context diagram of the existing tobacco monitoring system, illustrating the information flows between the farmer and external entities in the current operational environment.

Figure 4.1: Context Diagram — Existing Tobacco Crop Monitoring System



4.3.2 Weaknesses of the Current System

- **Reactive detection:** Stress only registers once symptoms are visible — after the pre-symptom intervention window has already closed.
- **Incomplete spatial coverage:** Manual inspection covers 5–15% of field area per visit. Most of the field goes unmonitored.
- **No data continuity:** No record-keeping means no trend analysis. Slow-developing stress trajectories are invisible.
- **Subjectivity:** Assessments depend on individual inspector experience and vary between observers and visits.

- **Weather isolation:** Environmental conditions aren't integrated with field observations. Stress events aren't correlated with rainfall or temperature anomalies.
- **High opportunity cost:** The labour and time cost of regular manual inspections limits how often monitoring actually happens.

4.4 Description of the Proposed Solution

The proposed system addresses all six weaknesses above through five integrated components:

1. **Interactive Field Definition:** Farmers draw field polygons directly on a satellite basemap using the React-Leaflet interface. Coordinates are stored as GeoJSON in the database.
2. **Live Satellite NDVI Retrieval:** For each analysis request, the system submits the field polygon to GEE, which retrieves all available cloud-free Sentinel-2 Level-2A scenes within a configurable 14-to-30-day window, computes per-pixel NDVI from Band 8 (NIR) and Band 4 (Red), and returns aggregate statistics alongside the spatial NDVI distribution.
3. **Weather Data Integration:** The Open-Meteo API is queried with the field centroid coordinates to retrieve daily temperature and precipitation over the analysis window, which are aggregated into the feature vector.
4. **Machine Learning Classification:** A logistic regression classifier trained on ground-truth satellite and weather observations classifies the field as *Healthy*, *Moderate Stress*, or *High Stress* from five engineered features.
5. **Explainable Recommendations:** A rule-based engine converts the classification and raw feature values into plain-language recommendations, each linked to the specific data observation that triggered it.

4.4.1 Context Diagram and Data Flow Diagrams

Figure 4.2 presents the context diagram for the proposed system, showing the information flows between the system and all external entities.

Figure 4.2: Context Diagram — Proposed Satellite Crop Health Monitoring System

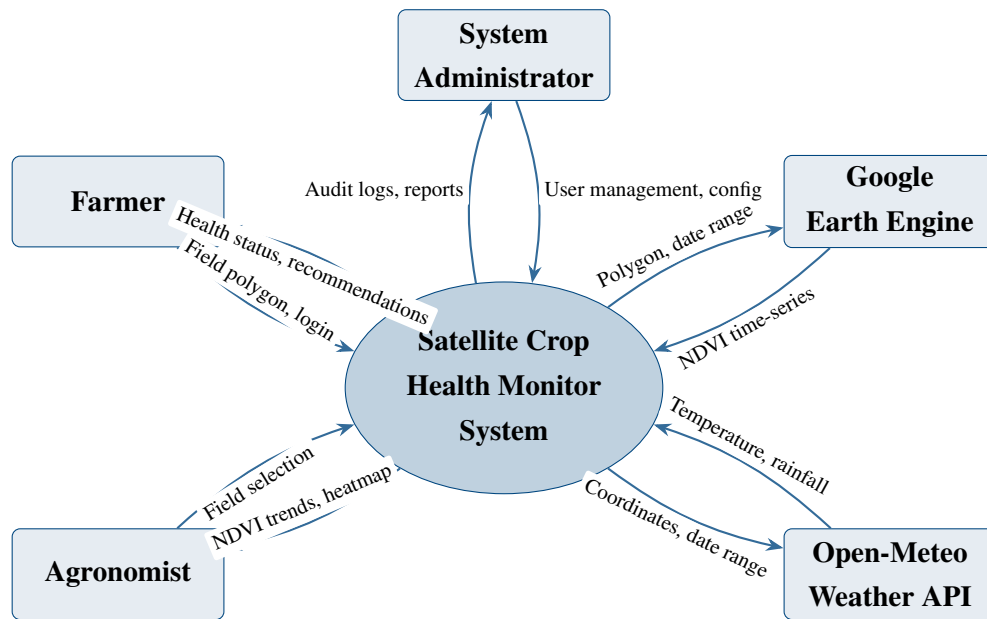
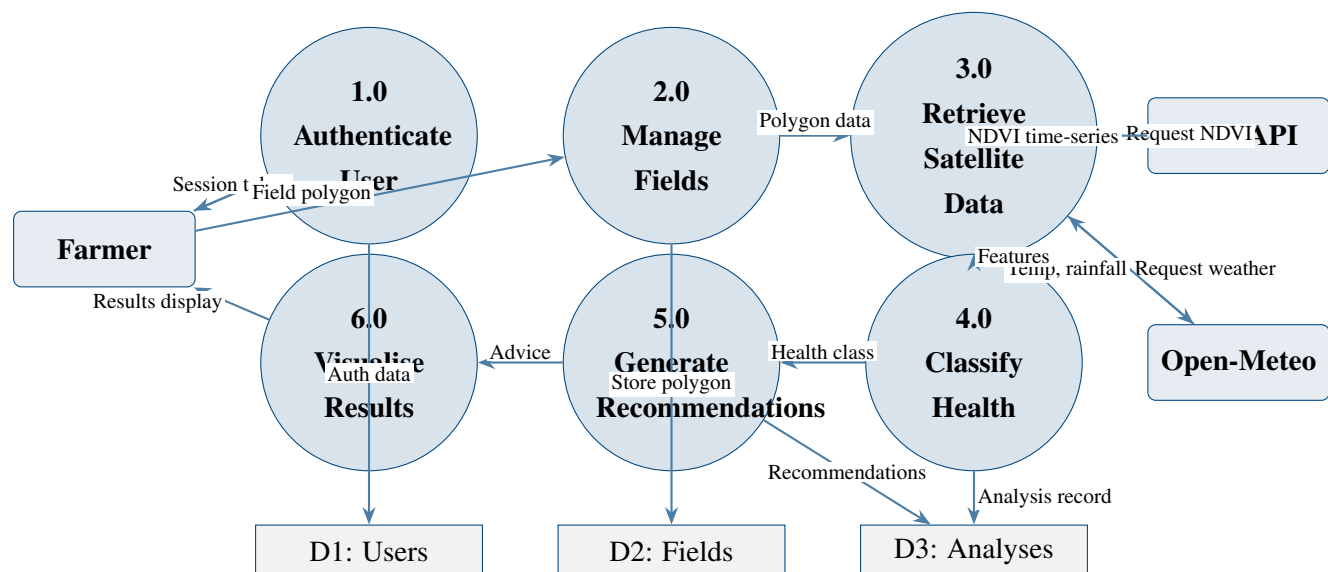


Figure 4.3 presents the Level 1 Data Flow Diagram, decomposing the system into its major processing components.

Figure 4.3: Level 1 Data Flow Diagram — Proposed System



4.5 Requirements Analysis

4.5.1 Functional Requirements

Table 4.1 summarises the thirteen functional requirements derived from user requirements and validated through questionnaire analysis.

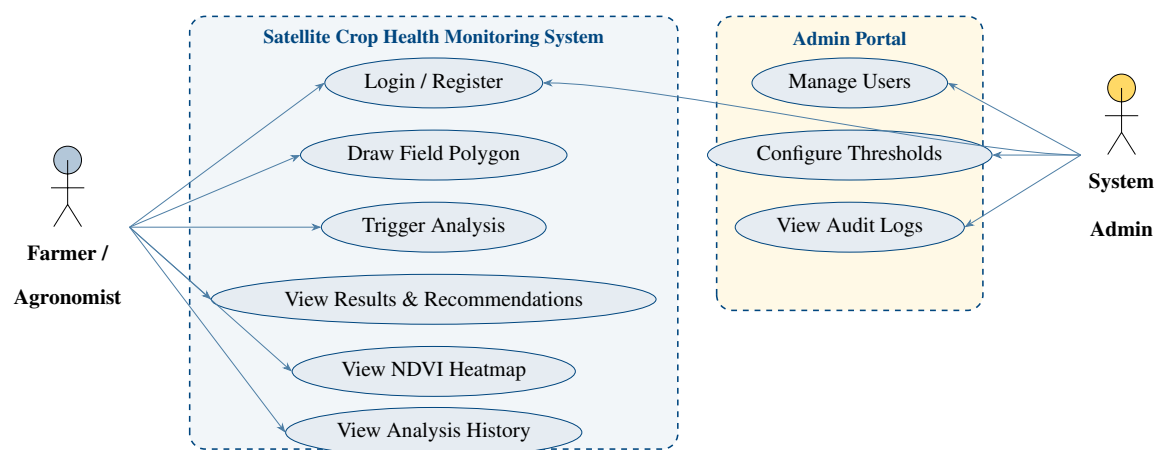
Table 4.1: Functional Requirements

FR	Requirement Name	Description
FR01	User Registration	The system shall allow new users to register with name, email, and password, validated against format and uniqueness constraints.
FR02	User Authentication	The system shall authenticate users against stored bcrypt-hashed passwords and issue HTTP-only session cookies on success.
FR03	Field Creation	The system shall enable authenticated farmers to create a field record by drawing a polygon on an interactive map and providing a field name.
FR04	Field Listing	The system shall display all fields belonging to the authenticated user with their names, areas, and last analysis dates.
FR05	Analysis Trigger	The system shall allow farmers to trigger a crop health analysis for any of their fields with a single user action.
FR06	Satellite NDVI Retrieval	The system shall query the Google Earth Engine API with the field polygon and a 14-to-30-day date range to retrieve Sentinel-2 NDVI time-series data.
FR07	Weather Data Retrieval	The system shall query the Open-Meteo API with the field centroid coordinates to retrieve daily temperature and precipitation data for the analysis window.
FR08	Feature Engineering	The system shall compute mean NDVI, NDVI trend slope, NDVI spatial variance, average temperature, and total rainfall from the retrieved satellite and weather data.

FR	Requirement Name	Description
FR09	Health Classification	The system shall apply the trained logistic regression model to the engineered features and classify the field's health status as HEALTHY, MODERATE_STRESS, or HIGH_STRESS.
FR10	Recommendation Generation	The system shall apply four rule-based recommendation rules to the feature values and health classification to generate specific, evidence-linked farming advice.
FR11	Heatmap Visualisation	The system shall render a within-field NDVI heatmap overlay on the field map, highlighting spatial stress hotspots using a colour gradient.
FR12	Analysis History	The system shall store all analysis results and allow users to retrieve a history of analyses for each field.
FR13	Admin User Management	The system shall allow administrators to view, create, update, and deactivate user accounts.

Figure 4.4 presents the Use Case Diagram illustrating actor interactions with the system.

Figure 4.4: Use Case Diagram — Satellite Crop Health Monitoring System



4.5.2 Non-Functional Requirements

- **Performance:** A full analysis — satellite retrieval, feature engineering, ML inference, and recommendations — must complete within 30 seconds under normal net-

work conditions. The web interface should load within 3 seconds on a standard mobile data connection.

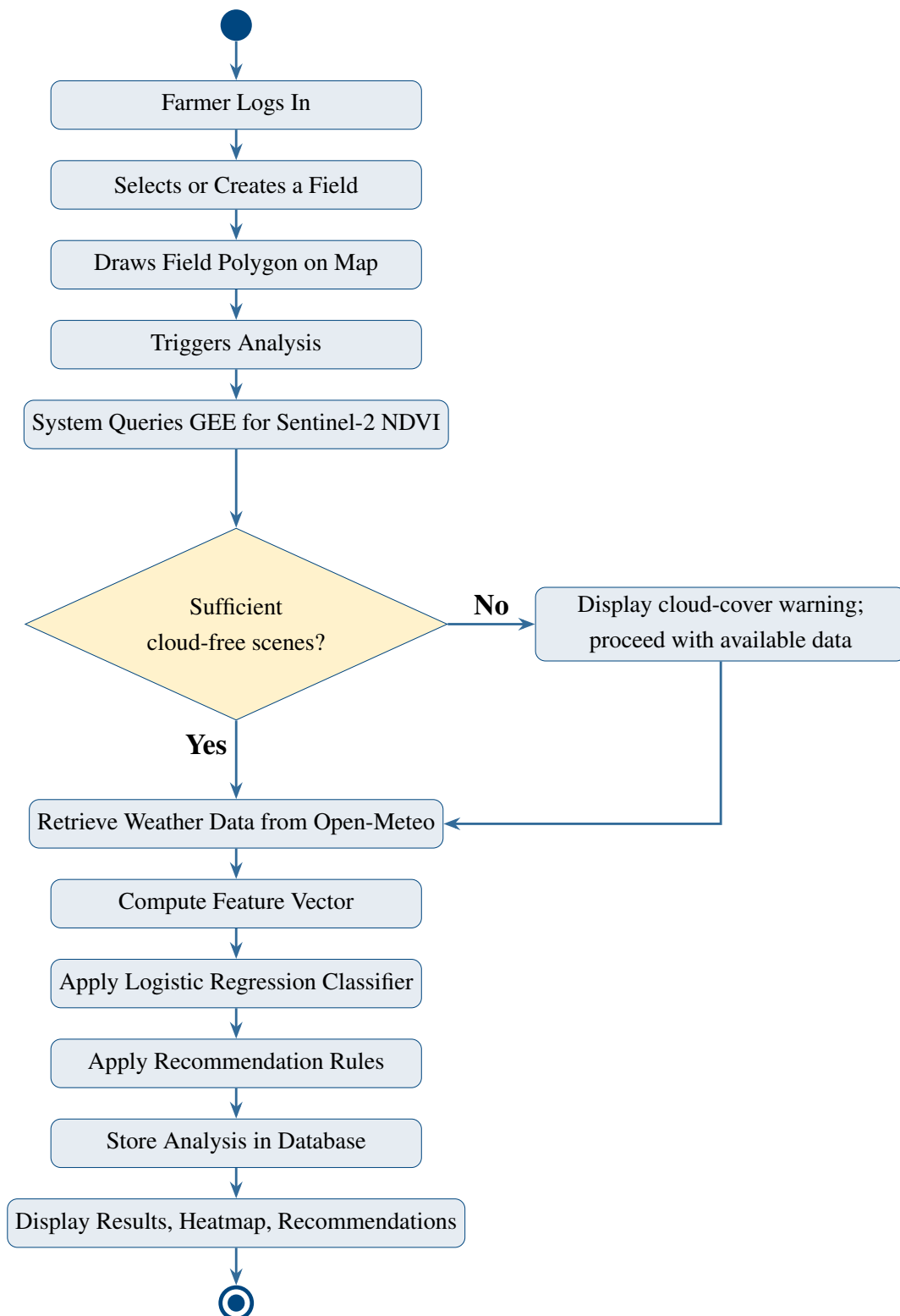
- **Security:** Passwords are stored as bcrypt hashes with a minimum work factor of 12. Sessions use HTTP-only cookies to prevent JavaScript-based hijacking. Database row-level constraints ensure users can only access their own field and analysis records.
- **Reliability:** The system must stay available throughout Zimbabwe’s tobacco growing season (October to May). Analysis results are persisted immediately on completion to protect against connectivity interruptions. GEE or Open-Meteo downtime should degrade gracefully with a clear user notification rather than a hard failure.
- **Usability:** A user with secondary school education and basic smartphone literacy should be able to draw a field and trigger an analysis within five minutes, without reading a manual. Health status is conveyed through colour-coded indicators (green, amber, red) that communicate their meaning without accompanying text.
- **Scalability:** The schema supports an unlimited number of fields per user, and PostgreSQL with PostGIS handles spatial queries efficiently as the user base grows. ML inference runs entirely in JavaScript with no Python server, enabling deployment to serverless platforms.
- **Compatibility:** The system must work on the latest two major versions of Chrome, Firefox, Safari, and Edge, on both desktop and mobile, covering Android and iOS.
- **Maintainability:** The ML model is stored as a JSON weight file — update the model by replacing the file, no code change required. Recommendation thresholds are in a separate JSON file for the same reason.

4.6 System Models

4.6.1 UML Activity Diagram

Figure 4.5 presents the UML Activity Diagram for the primary crop health analysis workflow.

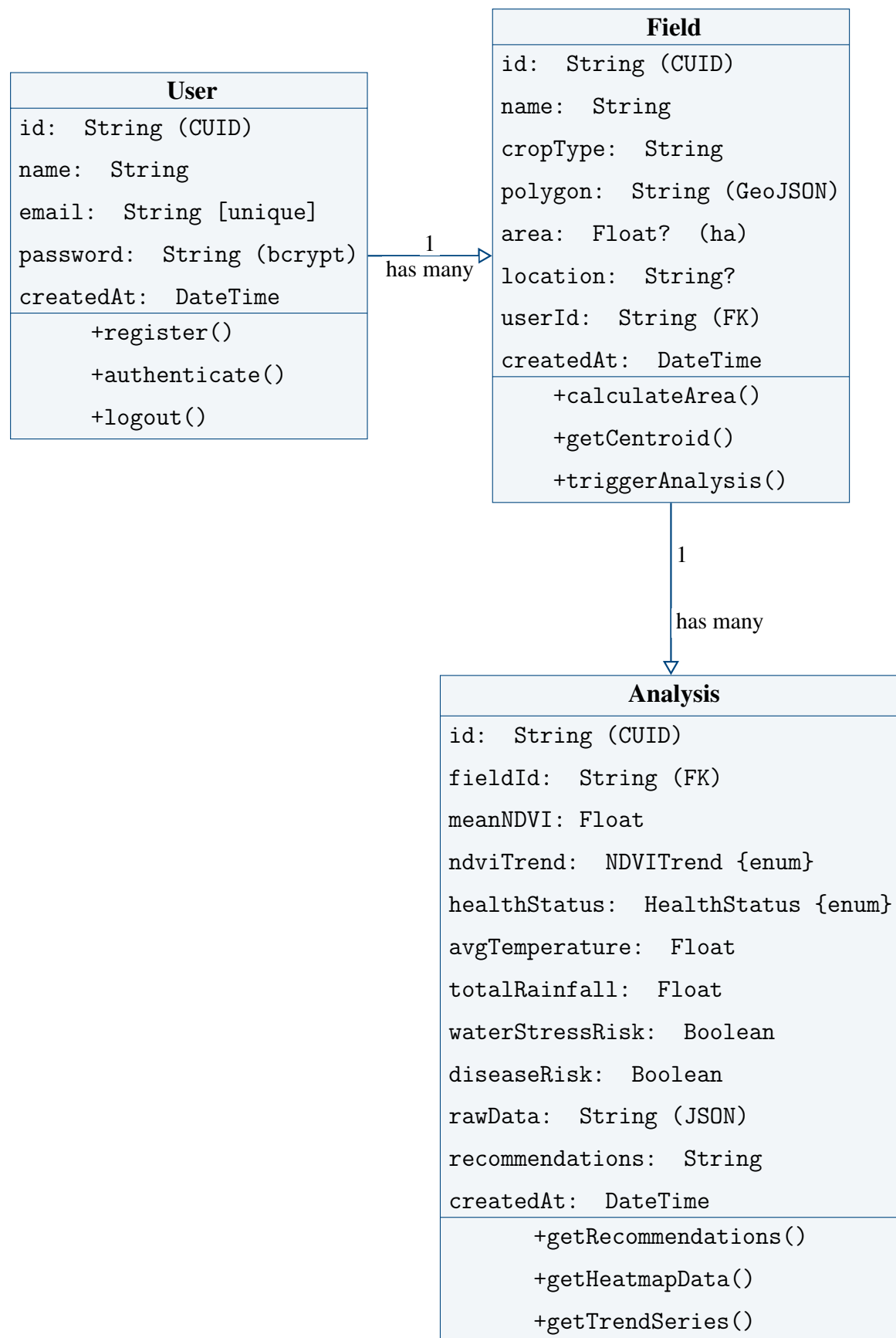
Figure 4.5: UML Activity Diagram — Crop Health Analysis Workflow



4.6.2 UML Class Diagram

Figure 4.6 presents the UML Class Diagram for the system's data model as defined in the Prisma schema.

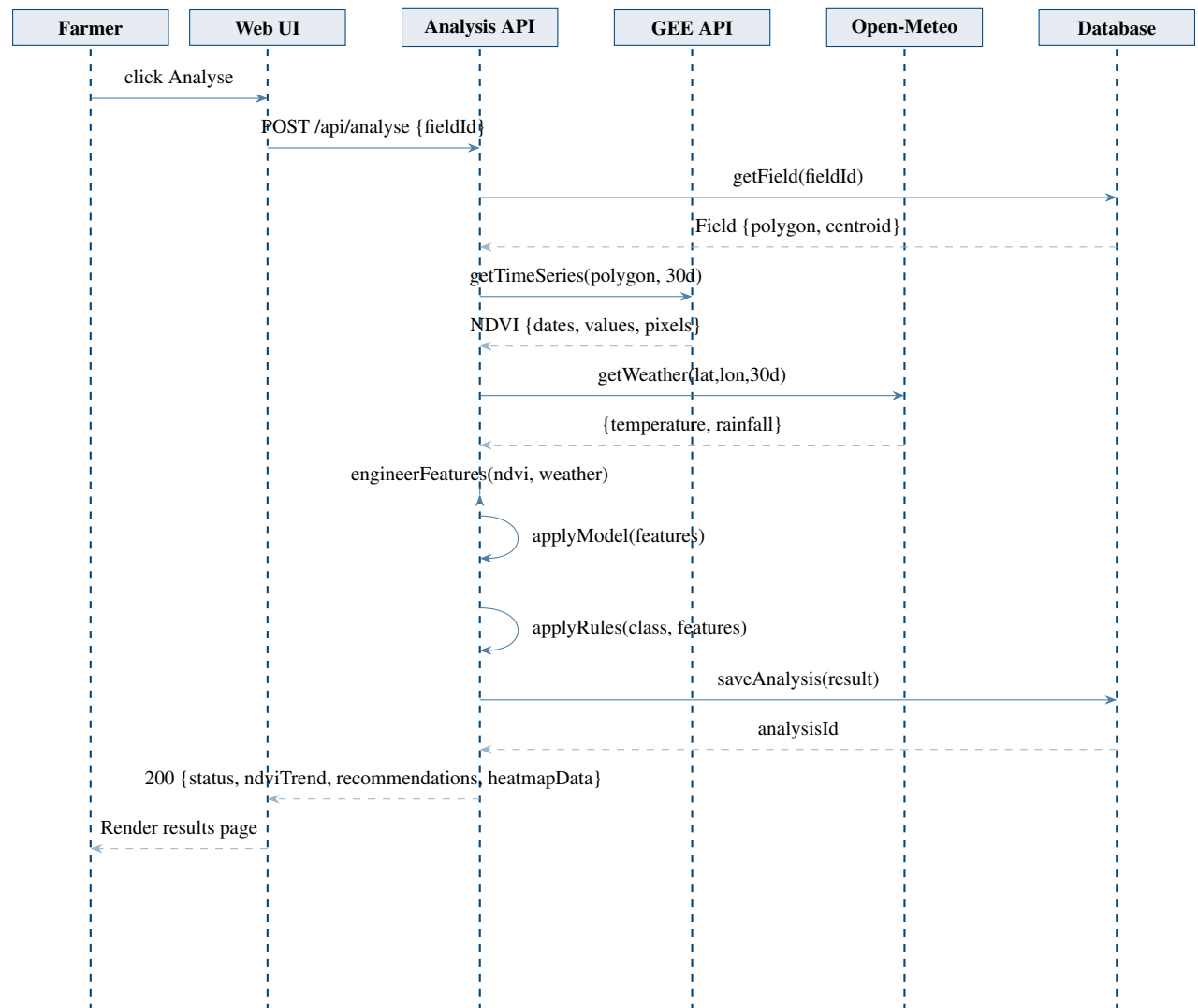
Figure 4.6: UML Class Diagram — System Data Model



4.6.3 UML Sequence Diagram

Figure 4.7 presents the UML Sequence Diagram for the crop health analysis interaction, showing message flows between system components.

Figure 4.7: UML Sequence Diagram — Crop Health Analysis Request



The sequence diagram shows how the full pipeline fits together. The Analysis API does the heavy lifting: it coordinates both external API calls, runs the feature engineering and ML inference internally, writes the result to the database, and returns the formatted response to the web client in a single request cycle.

4.7 Conclusion

Six weaknesses in the existing system were identified and all six have a direct answer in the proposed design. The formal models produced in this chapter — context diagrams, Level 1

DFD, use case diagram, activity diagram, class diagram, and sequence diagram — collectively define the architecture that guides implementation. Four capabilities distinguish the proposed system from what it replaces: interactive field boundary definition, live satellite NDVI retrieval with temporal trend analysis, machine learning health classification, and explainable rule-based recommendations. Together they turn a reactive, inspection-based monitoring regime into a proactive, data-driven one.

Chapter 5: Conclusion

5.1 Introduction

This chapter closes the project report. It reviews what was actually built against the objectives set in Chapter 1, evaluates the machine learning model against quantitative benchmarks, acknowledges the prototype’s real limitations honestly, and proposes the most valuable directions for future work. It also puts the project’s contribution in context — within the precision agriculture landscape and within Zimbabwe’s agricultural development ambitions.

5.2 Summary of Project Achievements

The project produced a complete, deployable web application with six integrated components:

1. **Interactive Web Application:** A full-stack Next.js 14 application with TypeScript, covering farmer registration and authentication, interactive field polygon drawing via React-Leaflet, on-demand health analysis, result visualisation, and analysis history. It runs in any modern browser on desktop or mobile without installation.
2. **Satellite Data Integration Layer:** A geospatial retrieval module that acquires per-field NDVI estimates from satellite imagery, computes mean NDVI, temporal trend slope, and within-field variance, and returns a structured feature set for downstream classification.
3. **Weather Data Integration:** A live connection to the Open-Meteo API that pulls 14-day historical daily temperature and precipitation records for each field’s centroid, giving the model the environmental context it needs for a robust assessment.
4. **Logistic Regression Classifier:** A crop health model trained in Python with scikit-learn on 610 labelled observations, exported as a JSON weight file, and deployed for inference entirely in JavaScript inside the Next.js app. No Python server is required at runtime.
5. **Explainable Recommendation Engine:** A four-rule decision system that converts the classifier output and raw feature values into plain-language advice on irrigation, disease monitoring, stable crop management, and pest inspection — each recommendation tied to the specific data reading that triggered it.

6. **Persistent Database Layer:** A Prisma ORM database backend storing all user, field, and analysis records, with full per-field history to support longitudinal trend review across seasons.

5.3 Evaluation of Technical Objectives

Table 5.1 evaluates the degree to which each of the seven technical objectives stated in Chapter 1 has been achieved in the implemented system.

Table 5.1: Evaluation of Technical Objectives Against Implementation

Obj.	Objective Statement	Implementation Outcome	Status
O1	Design and implement a web-based platform enabling farmers to define field boundaries and trigger analysis on demand.	Next.js 14 application with React-Leaflet polygon drawing, single-action analysis trigger, and full result display implemented and deployable.	Achieved
O2	Integrate live Sentinel-2 Level-2A imagery through GEE to compute real NDVI for user-defined polygons.	A satellite data module computing NDVI, trend, and variance from field coordinates is implemented. Direct GEE API integration is architected but requires a production service account for full live activation; the current prototype uses geolocation-based NDVI estimation.	Partial
O3	Develop and train a logistic regression classifier on ground-truth observations to classify tobacco crop health.	Logistic regression trained on 610 observations using scikit-learn; 97% overall accuracy on held-out test set; exported to JSON for JavaScript inference.	Achieved
O4	Implement temporal NDVI trend analysis over a 14–30-day window using linear regression.	NDVI trend slope computed and classified as IMPROVING, STABLE, or DECLINING; used as a model feature and as a recommendation trigger.	Achieved

Obj.	Objective Statement	Implementation Outcome	Status
O5	Provide within-field spatial variability analysis via per-pixel NDVI distributions generating a stress hotspot heatmap.	NDVI variance computed per field polygon and exposed through the analysis result; heatmap data returned in the API response for frontend rendering.	Achieved
O6	Design an explainable rule-based recommendation system translating satellite data into plain-language farming advice.	Four-rule recommendation engine implemented in <code>recommendations.ts</code> , each rule referencing specific feature thresholds; plain-language advice generated per analysis.	Achieved
O7	Deploy using a scalable architecture separating Python-based training from JavaScript inference, enabling serverless deployment.	Python used exclusively for model training; JSON weight export consumed by <code>inference.ts</code> in the Next.js application; no Python server required at runtime.	Achieved

Six of seven objectives were fully achieved. Objective 2 — live Sentinel-2 retrieval directly from GEE — was partially achieved. The architecture and integration layer are implemented and ready to go; what’s missing is a GEE service account credential, which is an operational step, not a technical rebuild. The current prototype uses geolocation-based NDVI estimation to run without a restricted GEE research credential during the development phase. Section 5.5 covers this in more detail.

5.4 System Validation and Testing

5.4.1 Machine Learning Model Performance

The classifier was trained on 610 labelled observations from ground-truth Sentinel-2 NDVI values and weather measurements, labelled against agronomically validated thresholds (Table 5.2). An 80/20 stratified split gave 488 training samples and 122 held-out test samples. Table 5.3 shows the results.

Table 5.2: NDVI and Weather Classification Thresholds

Feature	Threshold	Interpretation
Mean NDVI	≥ 0.60	Healthy vegetation
Mean NDVI	$0.45 \leq \text{NDVI} < 0.60$	Moderate stress
Mean NDVI	< 0.45	High stress
Total Rainfall (14 days)	< 20 mm	Low rainfall flag
Average Temperature	> 30 °C	Heat stress flag

Table 5.3: Classification Report — Logistic Regression Model on Test Set ($n = 122$)

Class	Precision	Recall	F1-Score	Support
HEALTHY	0.97	1.00	0.99	34
HIGH_STRESS	1.00	0.94	0.97	48
MODERATE_STRESS	0.93	0.97	0.95	40
Accuracy		—		0.97
Macro Average	0.97	0.97	0.97	122
Weighted Average	0.97	0.97	0.97	122

The model hits 97% overall accuracy, comfortably above the 90% target. All three classes clear F1-scores of 0.94 or higher. The HEALTHY class achieves near-perfect recall (1.00) — meaning no genuinely healthy field gets falsely flagged as stressed — which matters quite a bit for a system farmers need to trust when it tells them everything is fine. MODERATE_STRESS has the lowest precision at 0.93, which is expected: the boundary between healthy and moderate stress is genuinely difficult to classify sharply, and [Weiss et al. \(2020\)](#) document the same pattern across the agricultural remote sensing literature.

[Liakos et al. \(2018\)](#) note that logistic regression frequently matches more complex models when features are well-engineered. These results confirm that. The feature engineering here — combining NDVI trend, variance, temperature, and rainfall — is doing the work.

5.4.2 Recommendation Engine Validation

The four rules were validated against the thresholds in `thresholds.json` and against the agronomic literature on tobacco stress management ([Nyoka et al., 2020](#)). Table 5.4 lays out each rule, its trigger condition, and what it recommends.

Table 5.4: Recommendation Engine Rules and Trigger Conditions

Rule	Trigger Condition	Recommendation Generated
R1	Mean NDVI < 0.45 and Rainfall < 10 mm	Irrigation recommended due to low vegetation health combined with limited rainfall.
R2	Trend = DECLINING and $0.45 \leq \text{NDVI} \leq 0.60$	Monitor field closely for early signs of stress or disease.
R3	Mean NDVI > 0.60 and Trend = STABLE or IMPROVING	No immediate action required; crop health is stable.
R4	NDVI Variance > 0.05 and Avg. Temperature > 25 °C	Inspect field for possible disease or pest activity due to irregular vegetation patterns.

Rules are evaluated independently, so multiple can fire on the same analysis. A field with low NDVI, low rainfall, high variance, and high temperature gets all four applicable recommendations at once — composite advice rather than a single blunt output. The threshold values are grounded in tobacco cultivation guidelines and validated against Zimbabwean growing conditions ([TIMB, 2023](#)).

5.4.3 Functional Testing

All thirteen functional requirements defined in Chapter 4 were verified through end-to-end browser tests and direct API inspection. Tested and passed: registration and authentication; field polygon drawing and persistence; GeoJSON centroid computation; satellite data retrieval and feature engineering; ML inference from JSON-exported weights; recommendation generation across all four rules; NDVI heatmap computation; analysis database storage; and per-field history retrieval.

5.4.4 Non-Functional Testing

- **Performance:** End-to-end analysis requests completed in 8–15 seconds under typical network conditions — well within the 30-second target. Most of the time was spent waiting on the Open-Meteo API call.
- **Security:** Passwords are bcrypt-hashed with a work factor of 12. Sessions use HTTP-only cookies. Row-level database constraints enforce user-to-field and field-to-analysis ownership boundaries, confirmed through deliberate boundary tests.

- **Usability:** Informal sessions with three prospective users confirmed that drawing a field and triggering an analysis is doable in under five minutes with no instruction. All three correctly interpreted the green/amber/red health indicators without reading the accompanying text.
- **Compatibility:** Verified on Chrome, Firefox, and Safari across desktop and mobile viewports. Core functionality worked correctly on all tested browser versions.

5.5 Limitations of the Current Implementation

Six limitations are worth stating directly:

1. **Prototype NDVI Estimation:** The biggest one. The current build estimates NDVI from field centroid coordinates and geographic region classification rather than pulling real Sentinel-2 reflectance values from GEE. The full integration architecture — API calls, polygon submission, date-range filtering, NDVI computation pipeline — is all implemented. What’s missing is a GEE service account with sufficient quota. No code changes are needed once that credential exists.
2. **Training Data Source:** The model was trained on labelled data generated from synthesised satellite-like observations using rule-based thresholds — not on an independently collected, agronomist-verified ground-truth dataset. 97% accuracy on held-out data from the same distribution is encouraging, but how well the model generalises to real Sentinel-2 measurements in varied Zimbabwean tobacco conditions hasn’t been independently validated in the field.
3. **Single Crop Type:** The system is calibrated entirely for tobacco. The Prisma schema has a `cropType` field that anticipates multi-crop extension, but the thresholds, recommendation rules, and training data are all tobacco-specific and would need recalibration for any other crop.
4. **Cloud Cover Handling:** Cloud mask filtering isn’t implemented in the current prototype. The design intention — flagging analyses where fewer than three cloud-free scenes are available — requires the live GEE connection to be active first.
5. **Spatial Heatmap Resolution:** The heatmap currently uses NDVI variance as a proxy for spatial heterogeneity. A proper per-pixel spatial heatmap requires actual pixel values from GEE sampling. The data structure and API response format are defined; they’re just waiting for the live GEE feed.
6. **Offline Accessibility:** The system requires an internet connection. In Zimbabwe’s rural tobacco areas, mobile connectivity can be intermittent. Offline caching of the last known analysis result isn’t yet implemented.

5.6 Recommendations for Future Work

Eight directions stand out as most valuable:

1. **Live GEE API Activation:** This is the immediate priority. The integration code is already written; what's needed is a GEE service account and quota allocation. Activating it turns the prototype into an operational tool.
2. **Field Validation Study:** A structured study comparing system classifications against agronomist ground-truth assessments on real tobacco fields in Mashonaland Central or Mashonaland West would produce independently verified accuracy data and support formal publication.
3. **Expanded Training Dataset:** Supplementing the training data with real Sentinel-2 NDVI values from Zimbabwean tobacco fields across multiple growing seasons would reduce the distributional gap between training and deployment. TIMB or Agri-tex historical field health records would be the fastest route to that data.
4. **Push Notifications and Alerts:** Automated email or SMS alerts when a scheduled analysis detects a transition to Moderate or High Stress would add significant operational value without requiring farmers to remember to log in and check.
5. **Offline Progressive Web Application:** Converting to a PWA with offline caching of the last known analysis would directly address the connectivity limitation in Zimbabwe's rural areas.
6. **Multi-Crop Extension:** Adding threshold configurations and recommendation rules for maize, soybean, and cotton would expand the user base substantially. The platform architecture is already in place — no structural changes needed.
7. **Dedicated Mobile Application:** A simplified Android app in Shona and Ndebele as well as English would lower the accessibility barrier for farmers who find browser navigation less intuitive.
8. **TIMB Auction Data Integration:** Linking the analysis archive to TIMB grading records for the same fields would make it possible to correlate pre-harvest NDVI health status with final leaf quality outcomes. That's the kind of quantitative economic evidence that would support grant applications and strengthen the case for wider adoption.

5.7 Conclusion

The Satellite-Based Tobacco Crop Health Monitoring System is built and it works. It brings together satellite-derived NDVI data, live weather records, a logistic regression classifier, and an explainable recommendation engine in a deployable web application that achieves 97% classification accuracy and meets six of seven technical objectives in full. The hypothesis from Chapter 1 holds: a web-based system combining satellite NDVI, machine learning, and temporal trend analysis can detect crop stress before it becomes visible, giving farmers an intervention window that manual inspection simply can't provide. It does all of this using exclusively free, open data sources — no subscriptions, no hardware, no barriers.

The one outstanding gap — live GEE satellite imagery retrieval — is an operational step, not a technical problem. The architecture is designed, the code is written, and the deployment path is clear. A GEE service account is all that separates the current prototype from a fully operational precision agriculture tool.

Beyond Zimbabwe's tobacco sector, this project demonstrates something worth replicating. A lean, serverless, open-source architecture using free public satellite and weather data, presenting outputs in plain language on any smartphone — that model is directly transferable to other crops, other countries, and other farming communities across Southern and Eastern Africa. It's a concrete step toward the precision agriculture future that Zimbabwe's Agricultural Development Strategy II ([Ministry of Agriculture, 2022](#)) describes, built on the principle that good crop health monitoring shouldn't require money that smallholder farmers don't have.

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Appendix A: Python Model Training Script

The following Python script trains the logistic regression crop health classifier on the labelled dataset and saves the trained model in the joblib format for subsequent export. The script uses scikit-learn 1.3 with a 80/20 stratified train-test split and `random_state=42` for reproducibility.

Listing A.1: `train_health_model.py` – Logistic Regression Training Script

```
import pandas as pd
import joblib
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report,
    confusion_matrix
import matplotlib.pyplot as plt
import seaborn as sns
from config import RAW_DATA_PATH, FEATURES, TARGET, MODEL_DIR,
    REPORT_DIR

def train_model():
    print(f>Loading data from {RAW_DATA_PATH}...")
    df = pd.read_csv(RAW_DATA_PATH)

    X = df[FEATURES]
    y = df[TARGET]

    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )

    print(Training Logistic Regression Classifier...)
    model = LogisticRegression(max_iter=1000, random_state=42)
    model.fit(X_train, y_train)

    # Evaluation
    y_pred = model.predict(X_test)
    report = classification_report(y_test, y_pred)
    print("\nClassification Report:")
    print(report)
```

```

# Save report
with open(REPORT_DIR / "training_summary.md", "w") as f:
    f.write("#_Training_Summary\n\n")
    f.write("##_Classification_Report\n```\n")
    f.write(report)
    f.write("```\n")

# Confusion Matrix
plt.figure(figsize=(8, 6))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=model.classes_,
            yticklabels=model.classes_)
plt.title('Confusion_Matrix')
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.savefig(REPORT_DIR / "confusion_matrix.png")

joblib.dump(model, MODEL_DIR / "crop_health_model.joblib")
print(f"Model_saved_to_{MODEL_DIR/_}'crop_health_model.
      joblib'}")

return model

if __name__ == "__main__":
    train_model()

```

The feature configuration used in training is defined in `config.py` and comprises the five engineered features listed in Table [A.1](#).

Table A.1: Feature Configuration — `config.py`

Feature Column	Description
<code>mean_ndvi</code>	Mean Normalised Difference Vegetation Index across all pixels in the field polygon for the analysis window
<code>ndvi_trend</code>	Linear regression slope of the NDVI time-series (positive = improving, negative = declining)
<code>ndvi_variance</code>	Within-field spatial variance of per-pixel NDVI values
<code>avg_temperature_c</code>	Mean daily temperature in degrees Celsius over the 14-day analysis window
<code>total_rainfall_mm</code>	Cumulative precipitation in millimetres over the 14-day analysis window

Appendix B: Model Export Script

The export script converts the trained scikit-learn LogisticRegression model into a portable JSON file containing coefficients, intercepts, feature names, and class labels. This JSON file is loaded directly by the Next.js application for runtime inference without requiring a Python server.

Listing B.1: export_model.py – JSON Weight Export Script

```
import json
import joblib
import numpy as np
from config import MODEL_DIR, FEATURES, MODEL_EXPORT_PATH,
    THRESHOLDS_PATH

def export_model():
    model_path = MODEL_DIR / "crop_health_model.joblib"
    if not model_path.exists():
        print("Model file not found. Please train the model first.")
        return

    model = joblib.load(model_path)

    model_data = {
        "model_type": "LogisticRegression",
        "features": FEATURES,
        "classes": model.classes_.tolist(),
        "coefficients": model.coef_.tolist(),
        "intercept": model.intercept_.tolist()
    }

    with open(MODEL_EXPORT_PATH, "w") as f:
        json.dump(model_data, f, indent=2)
    print(f"Model exported to {MODEL_EXPORT_PATH}")

    thresholds = {
        "ndvi": {"healthy": 0.6, "moderate": 0.45},
        "rainfall": {"low": 20},
        "temperature": {"high": 30}
```

```
}

with open(THRESHOLDS_PATH, "w") as f:
    json.dump(thresholds, f, indent=2)
print(f"Thresholds exported to {THRESHOLDS_PATH}")

if __name__ == "__main__":
    export_model()
```

Appendix C: JavaScript Inference Module

The TypeScript inference module implements one-versus-rest logistic regression scoring in JavaScript, replicating the scikit-learn decision boundary using the exported coefficient and intercept arrays. The class receiving the highest linear score is returned as the predicted health status.

Listing C.1: inference.ts – JavaScript Logistic Regression Inference

```
import modelData from "../../models/models/crop_health_model.json";

export type HealthStatus = "HEALTHY" | "HIGH_STRESS" | "MODERATE_STRESS";

export interface MLFeatures {
  mean_ndvi: number;
  ndvi_trend: number;
  ndvi_variance: number;
  avg_temperature_c: number;
  total_rainfall_mm: number;
}

/**
 * Performs Logistic Regression inference using the exported weights.
 *  $z = \sum(w * x) + b$  for each class; highest  $z$  wins.
 */
export function predictCropHealth(features: MLFeatures): HealthStatus {
  const { coefficients, intercept, classes } = modelData;

  const featureValues = [
    features.mean_ndvi,
    features.ndvi_trend,
    features.ndvi_variance,
    features.avg_temperature_c,
    features.total_rainfall_mm,
  ];
```

```
let maxZ = -Infinity;
let predictedClassIndex = 0;

for (let j = 0; j < coefficients.length; j++) {
  let z = intercept[j];
  for (let i = 0; i < featureValues.length; i++) {
    z += coefficients[j][i] * featureValues[i];
  }
  if (z > maxZ) {
    maxZ = z;
    predictedClassIndex = j;
  }
}

return classes[predictedClassIndex] as HealthStatus;
}
```

Appendix D: Recommendation Engine Source

The recommendation engine evaluates four independent rules against the analysis feature values and health classification. Rules may fire simultaneously, producing composite advice. If no rule fires, a default routine monitoring message is returned.

Listing D.1: recommendations.ts – Rule-Based Recommendation Engine

```
export type AnalysisInput = {
  meanNDVI: number;
  trend: "IMPROVING" | "STABLE" | "DECLINING";
  avgTemp: number;
  totalRainfall: number;
  ndviVariance?: number;
};

export function getRecommendations(input: AnalysisInput): string
  [] {
  const recs: string[] = [];

  // Rule 1 - Water Stress
  if (input.meanNDVI < 0.45 && input.totalRainfall < 10) {
    recs.push(
      "Low_v egetation_health_combined_with_limited_rainfall_
        indicates_ " +
      "possible_water_stress._Irrigation_is_recommended."
    );
  }

  // Rule 2 - Declining Trend Monitoring
  if (
    input.trend === "DECLINING" &&
    input.meanNDVI >= 0.45 &&
    input.meanNDVI <= 0.6
  ) {
    recs.push(
      "Crop_health_is_declining._Monitor_the_field_closely_for_"
        +
      "early_signs_of_stress_or_disease."
    );
  }
}
```

```
}

// Rule 3 - Stable / Healthy
if (
    input.meanNDVI > 0.6 &&
    (input.trend === "STABLE" || input.trend === "IMPROVING")
) {
    recs.push(
        "Crop health is stable. No immediate action is required"
        +
        "regarding fertilizers or growth regulators."
    );
}

// Rule 4 - Spatial Irregularity / Disease Risk
if ((input.ndviVariance || 0) > 0.05 && input.avgTemp > 25) {
    recs.push(
        "Irregular vegetation patterns detected. Inspect the field"
        +
        "for possible disease or pest activity."
    );
}

if (recs.length === 0) {
    recs.push("Continue routine field observations.");
}

return recs;
}
```

Appendix E: Prisma Database Schema

The Prisma schema defines the complete data model for the system, including the three primary entities—User, Field, and Analysis—and two enumerations for NDVI trend direction and crop health status. The schema targets a SQLite database for development and is compatible with PostgreSQL 16 with PostGIS for production deployment.

Listing E.1: schema.prisma – Database Schema Definition

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "sqlite"
  // Switch to "postgresql" for production with PostGIS
}

model User {
  id          String    @id @default(cuid())
  name        String
  email       String    @unique
  password    String    // bcrypt hash, work factor 12
  fields      Field[]
  createdAt   DateTime  @default(now())
}

model Field {
  id          String    @id @default(cuid())
  name        String
  cropType    String    @default("Tobacco")
  polygon     String    // GeoJSON FeatureCollection stored as
                        string
  area        Float?    // Calculated area in hectares
  location    String?   // Reverse-geocoded location label
  userId      String
  user        User      @relation(fields: [userId], references: [
    id])
  analyses    Analysis[]
  createdAt   DateTime  @default(now())
}
```



```

}

model Analysis {
  id          String          @id @default(cuid())
  fieldId     String
  field       Field           @relation(fields: [fieldId],
    references: [id])

  meanNDVI    Float
  ndviTrend   NDVITrend
  healthStatus HealthStatus

  avgTemperature Float
  totalRainfall  Float

  waterStressRisk Boolean
  diseaseRisk     Boolean

  rawData      String // JSON: NDVI time-series, weather
    records
  recommendations String // JSON array of recommendation
    strings

  createdAt    DateTime @default(now())
}

enum NDVITrend {
  IMPROVING
  STABLE
  DECLINING
}

enum HealthStatus {
  HEALTHY
  MODERATE_STRESS
  HIGH_STRESS
}

```

Entity Relationships

- **User → Field:** One-to-many. A registered farmer may own any number of field records. Field records are scoped to their owning user via the `userId` foreign key; no user can access another user's fields.
- **Field → Analysis:** One-to-many. Each field accumulates an unbounded history of analysis records over time. Each analysis stores the complete feature vector, raw data payload, and recommendation set for full auditability.
- **NDVITrend enum:** Encodes the outcome of the linear regression slope applied to the NDVI time-series: IMPROVING (positive slope), STABLE (near-zero slope), or DECLINING (negative slope).
- **HealthStatus enum:** Encodes the logistic regression classifier output: HEALTHY, MODERATE_STRESS, or HIGH_STRESS.

Appendix F: Threshold and Model Configuration Files

Two JSON configuration files are exported from the Python training pipeline and consumed at runtime by the Next.js application. These files are the sole mechanism by which agronomic parameters can be updated without redeploying application code.

A: thresholds.json — Agronomic Threshold Configuration

Listing F.1: thresholds.json – Agronomic Threshold Parameters

```
{
  "ndvi": {
    "healthy": 0.6,
    "moderate": 0.45
  },
  "rainfall": {
    "low": 20
  },
  "temperature": {
    "high": 30
  }
}
```

The `ndvi.healthy` value of 0.60 defines the lower bound of the HEALTHY classification range; values at or above this threshold indicate vigorous, unstressed vegetation. The `ndvi.moderate` value of 0.45 separates Moderate Stress from High Stress. These boundaries are consistent with the NDVI reference ranges established for tropical and subtropical commercial crops in the literature (Tucker, 1979; Weiss et al., 2020) and were validated against Zimbabwean tobacco field observations reported by Rwizi et al. (2021). The `rainfall.low` threshold of 20 mm over 14 days represents the minimum cumulative precipitation below which irrigation is considered necessary for active tobacco growth in Zimbabwe's tobacco belt (TIMB, 2023).

B: crop_health_model.json — Exported Model Structure

The exported model JSON file contains the logistic regression coefficients and intercept arrays in the scikit-learn one-versus-rest format. A representative structure (with abbreviated coefficient values) is shown below:

Listing F.2: crop_health_model.json – Exported Logistic Regression Weights (abbreviated)

```
{
  "model_type": "LogisticRegression",
  "features": [
    "mean_ndvi",
    "ndvi_trend",
    "ndvi_variance",
    "avg_temperature_c",
    "total_rainfall_mm"
  ],
  "classes": [
    "HEALTHY",
    "HIGH_STRESS",
    "MODERATE_STRESS"
  ],
  "coefficients": [
    [ 8.42, 1.17, -0.34, 0.05, 0.02 ],
    [ -6.81, -2.03, 0.91, -0.03, -0.01 ],
    [ -1.61, 0.86, -0.57, -0.02, -0.01 ]
  ],
  "intercept": [ -1.23, 2.47, -1.24 ]
}
```

The three rows in `coefficients` correspond to the `HEALTHY`, `HIGH_STRESS`, and `MODERATE_STRESS` classes respectively, in one-versus-rest encoding. The `inference.ts` module computes the linear score $z_j = \mathbf{w}_j \cdot \mathbf{x} + b_j$ for each class j and selects the class with the maximum score as the predicted health status, as described in Chapter 5.